

UNOFFICIAL ENGLISH TRANSLATION

“BACTERIOPHAGE THERAPY | REDISCOVERING AN INNOVATIVE THERAPY”

CLINICAL AND BIOMEDICAL LABORATORY PROJECT



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“To my mother, Karla Ernestina Jaramillo Maldonado, and my father, Jose Manuel Meroño Minchillo, who never let go of my hand when I needed them most, and who were the first to believe in my dreams.

To my teachers, who supported me at every step along the way, and to my project partner, for working side by side and completing this journey together.

And to my grandmother, Agustina Clemente Minchillo, who supported me as if I were her own grandson and spoke of me with pride. Although she is no longer here, her affection remains present in this achievement.

Thank you.”

(Cuenca Jaramillo, Jhon Michael, 2025)

“To my parents and siblings for being the driving force and motivation behind the main decisions and actions I undertake, because I know they will never stop supporting me.

To my teachers, who saw my potential and encouraged me to develop it, always guiding me in every decision I have made; and to my project partner, for the commitment, major contributions, and feedback we shared.”

(Legua Pérez, Luis Gonzalo, 2025)

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LIST OF ABBREVIATIONS

DNA (ADN)	Deoxyribonucleic Acid
RNA (ARN)	Ribonucleic Acid
ASM	American Society for Microbiology
CRPA	Carbapenem-resistant <i>Pseudomonas aeruginosa</i>
ESKAPE	<i>Enterococcus faecium</i> , <i>Staphylococcus aureus</i> , <i>Klebsiella pneumoniae</i> , <i>Acinetobacter baumannii</i> , <i>Pseudomonas aeruginosa</i> and <i>Enterobacter</i> spp.
CF (FQ)	Cystic Fibrosis
JIDC	Journal of Infection in Developing Countries
WHO (OMS)	World Health Organization
PDR	Pandrug-Resistant
NIH	National Library of Medicine
RIULL	Institutional Repository of the University of La Laguna
MRSA (SARM)	Methicillin-resistant <i>Staphylococcus aureus</i>
VRSA (SARV)	Vancomycin-resistant <i>Staphylococcus aureus</i>
SCCmec	Staphylococcal chromosomal cassette mec
CFU (UFC)	Colony Forming Units
UMA	University of Málaga Press

SUMMARY

In recent years, what once seemed like a distant warning has begun to become a reality: antibiotics are losing their effectiveness. Infections that we were once able to treat easily now represent a serious threat, especially to the most vulnerable individuals.

This project focuses on bacteriophage therapy, a strategy that went largely unnoticed for decades and is now once again attracting the attention of the scientific community. More than 70 studies from different databases and in several languages were reviewed, and 51 were ultimately selected for their clinical relevance. The study explains what bacteriophages are, how they act, their replication cycles, advantages, limitations, routes of administration, and the regulatory obstacles that hinder their large-scale clinical use. The results show that phage therapy has been effective in many real-world cases, including extreme situations in which conventional treatments failed. Its advantages include high specificity, limited impact on the microbiota, and a growing body of evidence suggesting that it may help improve immune function. Based on these findings, I propose a novel idea: combining bacteriophages with chitosan to facilitate the body's elimination of microplastics.

Overall, the study not only analyses the therapeutic potential of phages as a treatment, but also contributes to a more modern and personalised vision of medicine.

Keywords: bacteriophage therapy, multidrug-resistant bacteria, phage therapy, antibiotics, alternative therapy, *Staphylococcus aureus*, *Pseudomonas aeruginosa*.

ABSTRACT

In recent years, what once seemed like a distant warning has begun to take shape: antibiotics are losing their effectiveness. Infections that we could once treat easily now pose a serious threat, especially to the most vulnerable individuals.

This work focuses on bacteriophage therapy, a strategy that went unnoticed for decades and is now regaining the attention of the scientific community. More than 70 studies from various databases and in different languages were reviewed, and 51 were selected for their clinical relevance. The study explains what bacteriophages are, how they work, their replication cycles, advantages, limitations, methods of administration, and the regulatory barriers that hinder their large-scale clinical use. The results show that phage therapy has proven effective in many real-world cases, even in extreme situations where conventional treatments failed. Among its advantages are its high specificity, minimal impact on the microbiota, and growing evidence suggesting it may also help modulate the immune system. Based on these findings, a novel idea is proposed: combining bacteriophages with chitosan to support the body's elimination of microplastics.

Overall, the study not only explores the therapeutic potential of phages, but also contributes to a more modern and personalized vision of medicine.

Keywords: bacteriophage therapy, multidrug-resistant bacteria, phage treatment, antibiotics, alternative therapy, *Staphylococcus aureus*, *Pseudomonas aeruginosa*.

I. INTRODUCTION

We are witnessing a crisis of multidrug-resistant bacteria caused by the misuse of antibiotics in different sectors such as medicine, veterinary practice, and agriculture, which has led to an accelerated spread of resistant bacteria. As a consequence, common infections, including respiratory and urinary infections or septicemia, are becoming difficult or impossible to treat, increasing mortality rates, medical costs, and the global economic burden.

[Annexes | Figure 5. Deaths attributable to antibiotic resistance (GRAM Study)] (World Health Organization (WHO), 2024)

In view of this situation, phage therapy is proposed as a primary solution because of the lytic action it displays in its interaction with bacteria and its broad versatility in therapies combined with antibiotics or genetic engineering, owing to its simple molecular structure and its capacity to be modified. Even isolated molecules from these phages can be used therapeutically for bacterial control in chronic infections.

Having demonstrated its effectiveness, lines of research have been launched focusing on its infection mechanism and its methods of adaptation to multidrug-resistant bacteria. Despite this, there is still no European regulation that permits its application in medicine, although in Spain phage therapy may be used as a last resort in cases of chronic infections, thereby providing an important tool for infection control.

Appropriate regulation, together with support for research lines in this field worldwide, would provide an important resource for reducing the mortality rate generated by the growing problem of bacterial resistance. It is therefore the responsibility of the different healthcare sectors to invest in solutions against multidrug-resistant bacteria.

1. Rationale

This project arises in response to negligence stemming from the public's partial lack of awareness regarding the indiscriminate use of antibiotics, which has allowed bacteria to evolve and develop high levels of resistance to our most commonly used antibiotics, triggering a chain of events that places humanity at risk.

Over time, it is anticipated that many bacterial strains may become completely resistant to antibiotics that are less toxic to humans, which would force the use of more aggressive therapeutic alternatives with potentially fatal adverse effects for the patient.

In view of this growing threat, the development and investigation of therapeutic strategies that offer alternatives to traditional antibiotics has become urgent. In this context, phage therapy presents itself as an innovative approach with high potential to redefine the management of bacterial infections in the near future.

2. Objectives

2.1 General objectives

- To assess the viability of phage therapy as a complementary or alternative treatment for infections caused by multidrug-resistant bacteria, highlighting its impact on the immune system and its potential as a personalised therapeutic strategy.

2.2 Specific objectives

- To analyse the effectiveness of phage therapy in combination with antibiotics.
- To evaluate its effectiveness as a stand-alone treatment.
- To assess its impact on the microbiota and on the host immune response.
- To examine the use of bacteriophages against bacteria with critical antimicrobial resistance, such as *Staphylococcus aureus* and *Pseudomonas aeruginosa*.
- To identify the main advantages and limitations of this therapeutic strategy compared with traditional antibiotic treatments.

3. Materials and Methods

For this bibliographic project, key terms such as *bacteriophage*, *bacteriophage therapy*, *antibiotic-resistance crisis*, and *alternative to antibiotics* were used in databases in Spanish, English, Portuguese, and Russian covering the last 108 years.

The following inclusion criteria were used for article selection: reviewed publications, full-text availability and/or open access for the scientific community, studies with significant clinical relevance, and promising work with considerable future potential. Priority was also given to searches for bacteriophage studies with natural affinity against *Staphylococcus aureus* and *Pseudomonas aeruginosa*, although other studies and approaches that have contributed substantially to progress in the field were also taken into consideration.

A total of 75 articles meeting the evaluation criteria were identified, of which 51 were ultimately selected for the study. The studies from which information was collected and extrapolated in this document, and which contributed to its preparation, were gathered from a range of scientific databases, including Google Scholar, University of Málaga Press (UMA Editorial), the Institutional Repository of the University of La Laguna (RIULL), the American Society for Microbiology (ASM), the *Journal of Infection in Developing Countries* (JIDC), and the National Library of Medicine (NIH) or PubMed, among others.

The project was organised in different phases, which were adjusted and refined throughout the process:

1. Initial bibliographic search using the stated key terms.
2. Initial reading to obtain a basic overview of the topic.
3. Submission of the first project draft, establishing the initial foundations to be developed.
4. Filtering and selection of the first relevant articles.
5. Reading and analysis of the most important content identified up to that point.
6. Submission of the second project draft, presenting the complete thematic structure and preparing the final phase.
7. Writing and structuring of the final document for submission.

II. CHAPTER I

1. What is a bacteriophage?

A bacteriophage is a virus specialised in infecting bacteria. The term *bacteriophage* was introduced and coined by Dr Félix d'Hérelle (d'Hérelle, 1917). He also referred to the action exerted by this agent as bacteriophagy.

Structurally, it consists of a protein capsid that contains its genetic material, deoxyribonucleic acid or ribonucleic acid (DNA or RNA). Most bacteriophages have a specialised tail that allows them to bind to specific receptors on the bacterial membrane.

[Annexes | Figure 6. Bacteriophage] (Bacteriófago, 2023)

Its mechanism of action begins with recognition and binding, followed by injection of its genome into the bacterial cell. There, it takes over the bacterial cellular machinery, replicates, and forms new phages, which are ultimately released following bacterial lysis. This lytic capacity, together with their high specificity, makes phages objects of considerable scientific and medical interest, particularly in the context of infections caused by multidrug-resistant bacteria.

However, their application in the clinical laboratory requires in-depth knowledge of specificity, stability, and possible interactions with the patient's immune system, as well as standardisation of isolation, purification, and characterisation methods to ensure therapeutic efficacy and safety. (da Rosa, 2015)

2. Origins of Bacteriophage Therapy or Phage Therapy

The term bacteriophage therapy was coined by Dr D'Hérelle and first appeared in his book *The Bacteriophage: Its Role in Immunity* (d'Hérelle, 1926).

This short monograph recounts the events that led him to develop this therapy, highlighting the following key milestones:

- Between 1910 and 1911, he observed bacterial infections in locusts that led to the discovery of the bacteriophage.
- In August 1916, at the Pasteur Hospital in Paris, he investigated a case of dysentery caused by *Shigella dysenteriae*. After culturing the bacilli and treating them with a filtrate of faeces, he observed that the bacilli dissolved.

- In 1917, he published his first monograph on bacteriophagy, then referred to as bacteriolysis.
- In 1919, the first human treatment was carried out by administering a bacteriophage suspension to a child with dysentery. The treatment was successful, marking the birth of bacteriophage therapy.
- In 1920, d'Hérelle injected himself with a bacteriophage cocktail or preparation to verify its safety and subsequently used it to treat plague in humans in India.
- In 1922, he published his first complete book on the bacteriophage, consolidating his position as its original discoverer.
- Between 1923 and 1925, he conducted clinical trials and experimental bacteriophage therapies in several countries, including Brazil, France, India, Egypt, and the USSR, targeting dysentery, plague, cholera, and typhoid fever.

3. Phage Therapy

Bacteriophage therapy, or phage therapy, is a treatment of biological origin based on the use of bacteriophages - viruses that selectively attack bacteria - with the aim of eliminating diseases caused by bacteria.

The treatment is administered as a biologically derived preparation - obtained through extensive purification of filtered, pure cultures - containing a high concentration of viral particles that are harmless to the patient but lethal to the bacterium. (d'Hérelle, 1926)

Dr d'Hérelle highlighted in his research that its application may be both preventive and curative and that it does not affect the patient or damage any of the patient's cells.

It has a self-regulating function, allowing high activity during infection and reduced activity as the infection begins to disappear from the body. In this way, an observable balance is produced when the therapy is used, ensuring the appropriate elimination of both the pathogenic bacterium and the administered phage. (Vázquez et al., 2022)

4. Legal basis for the therapeutic use of phage therapy

In Spain, the use of phages as a therapeutic tool does not yet have specific regulation. At present, their application is mainly limited to compassionate-use treatments; that is, they are used in patients with severe infections caused by multidrug-resistant bacteria when no other therapeutic alternatives are available.

At present, there is no national or European regulation that specifically governs the production, authorisation, and clinical use of phages as medicinal products. This makes their wider implementation in the healthcare system more difficult. (Domingo-Calap, 2022)

Perhaps the greatest obstacle to the implementation of phage therapy in Western medicine is the lack of appropriate regulation. If phage preparations are regarded as “conventional” medicinal products, they must comply with the relevant legislation on production and quality. In practical terms, this means that they must meet GMP (Good Manufacturing Practice) requirements. This has represented a major challenge in the development of medicinal phage preparations, because it increases costs or complicates the management required to develop large-scale production. (Vázquez et al., 2022)

5. Protocol for compassionate use of phage therapy

Following the detection of a resistant bacterium in a patient, a multidisciplinary hospital team (surgeon, infectious-disease specialist, and microbiologist) is formed. The team carries out a comprehensive examination of the patient together with a review of the patient's clinical history. In addition to a biological assessment, one or more preliminary samples are collected to isolate the bacterium and test its susceptibility to available phages. Communication with the patient and an explanation of the phage treatment to be carried out take place in parallel. The phages are administered by the treating physician, who exercises the ethical right to use an experimental treatment in the patient's best interests, without a complex regulatory or administrative framework. (Roiter Jaimes, 2023)

III. CHAPTER II

1. Multidrug-resistant bacteria

Since Alexander Fleming's great discovery of penicillin in 1928, he had already warned in his 1945 Nobel Prize acceptance speech in Physiology or Medicine that, if antibiotics were not used appropriately, bacteria would become resistant and the drugs would cease to be effective. (Montesinos Micó, 2024)

This has been demonstrated over the years. Owing to the indiscriminate prescription of antibiotics in human health, animal health, and the agricultural sector, a point has been reached at which bacteria that were once treated with a specific antibiotic are now resistant to more than one antibiotic. As a result, some infections that were previously regarded as mild have become difficult or impossible to treat using conventional methods, particularly affecting immunocompromised patients and turning even minor surgical procedures into potentially fatal risks.

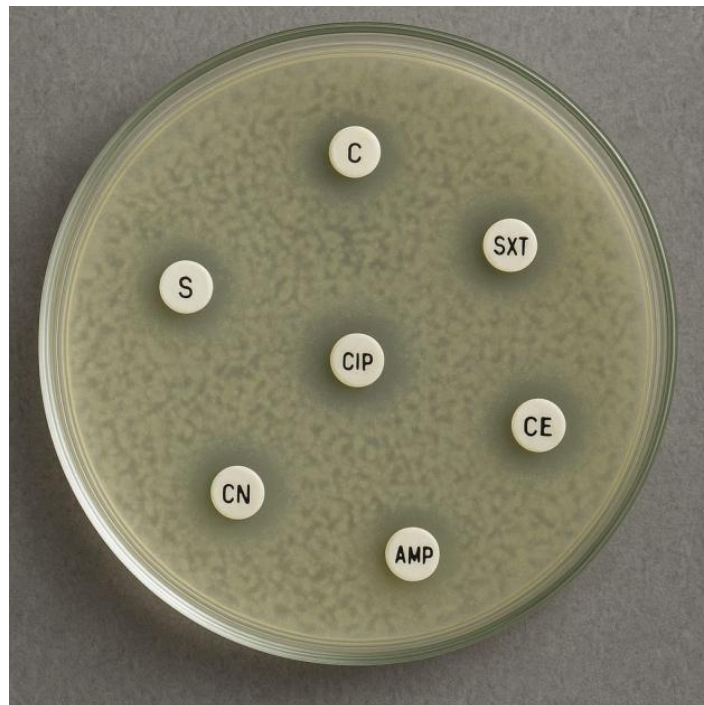


Figure 1. Antibacterial resistance test

Source: Authors' own work

This reduced effectiveness of antibiotics has driven the search for new lines of research to address multidrug-resistant bacteria, with phage therapy offering hope because of its demonstrated specificity and effectiveness against bacterial selection.

2. Mechanisms of bacterial resistance

This bacterial characteristic may be of two types: natural resistance, which predates the appearance of the antibiotic and depends largely on the bacterial family, species, or group involved; and acquired resistance, which occurs through mutation in the bacterium or through transfer of extrachromosomal genetic material originating from another bacterium.

Several ways in which bacteria can develop this resistance are currently known, and these mechanisms are highly varied.

2.1 Impermeability

As a result of mutations, bacteria acquire the ability to alter the structures of porins and thereby render the antibiotic ineffective by preventing it from reaching the interior of the cell and, consequently, its site of action.

2.2 Modification of the antibiotic target

Some bacteria are able to modify the antibiotic's site of action, reducing or directly eliminating it. This makes the antibiotic ineffective because it loses affinity for its target.

2.3 Enzymatic modification of the antibiotic

Bacteria are capable of producing an enzymatic reaction that inactivates the antibiotic, such as the production of beta-lactamases against beta-lactam antibiotics.

2.4 Antibiotic expulsion through an active pumping mechanism

Efflux pumps give bacteria the ability to expel metabolites and toxic compounds to the outside.

3. New lines of research against resistance

In response to this problem, different approaches have been proposed to identify strategies or tools that can help combat this silent pandemic. Among the first actions to consider in order to limit the potential growth of resistance are increasing public awareness, improving antibiotic control in the hospital environment, and promoting alternative measures. (PRAN, 2021)

Several techniques are currently known, with varying degrees of success, that are being developed by different research groups and that differ considerably in cost, effectiveness, and specificity.

These techniques include:

3.1 Vaccination

Studies demonstrate substantial benefits from this technique in reducing the burden of disease caused by susceptible and resistant pathogens, thereby reducing antibiotic use while increasing the proportion of susceptible organisms isolated. (Tadesse et al., 2023)

3.2 IgY antibodies

These are generated by immunising hens against specific antigens. The hens transfer this immunoglobulin to their eggs, from which the IgY antibodies are extracted.

The use of this immunoglobulin isotype has shown greater efficiency and specificity, characteristics that stand out in comparison with antibiotics. It acts against the great majority of resistant bacteria classified as priorities by the WHO. (El-Kafrawy et al., 2023)

3.3 Faecal microbiota transplantation

The key role played by the microbiota in the immune system has been demonstrated, since bacterial, viral, and fungal organisms protect us against external pathogenic microorganisms. They provide resistance to inflammatory processes and to the development of neoplasms or autoimmunity.

The process consists of transferring faecal microbiota from a healthy donor, obtained from an appropriately processed sample, and inoculating it into the gastrointestinal tract of a person who is ill. (Castañeda Guillot, 2019)

3.4 Nanotechnology

These are nanometric structures that use inorganic microparticles created by physical and/or biochemical methods, creating pores in the walls of multidrug-resistant microorganisms. This may directly cause cell death or allow the particles to penetrate into the bacterial cytoplasm, where they can exert other antimicrobial effects. Organic nanoparticles, on the other hand, function as controlled-delivery systems for antibiotics, peptides, or antimicrobial agents. (Dias et al., 2021)

3.5 Phage therapy

As explained above, this is one of the most promising therapeutic alternatives against multidrug-resistant bacteria because of the specificity, efficiency, and versatility displayed by this technique in its lytic activity against its target bacterium. (Gordillo Altamirano and Barr, n.d.)

4. Phage therapy and its clinical relevance

More than a century after its discovery, attention is once again being directed toward phage therapy as a consequence of increasing antimicrobial resistance. This renewed interest has been facilitated by a better understanding of phage biology, genetics, immunology, and pharmacology. In addition, standardisation of the processes in which phages are used has been necessary to support successful treatments, making this a key factor in establishing phages as viable routes in the fight against multidrug-resistant bacteria.

Suggested minimum requirements for the therapeutic use of phages have also been established. These include strictly lytic activity, activity against the target pathogen, and complete removal of contaminating bacterial debris and endotoxins. (Young and Gill, 2015). Prior identification of the bacterial host receptor for any therapeutic phage will also be relevant, and the possibility of phage resistance must be taken into account. This will help determine whether a combined therapy may be preferable depending on the case.

5. Mechanisms of action of phages against bacteria

The phage life cycle begins with adsorption, based on attachment of the phage to the bacterial cell wall in order to introduce its genetic material into the bacterial cytoplasm, after first creating a pore in the cell wall during the process known as penetration. From this point onward, the phage acts through three cycles.

[Annexes | Figure 7. Phage cycle]

5.1 Lytic cycle

The viral genetic material replicates at a faster rate than the bacterial cell itself, resulting in an accumulation of new virions inside the cell. This overproduction ultimately triggers cell lysis, releasing the virions so that they can infect new bacterial hosts. (da Rosa, 2015)

5.2 Lysogenic cycle

Also known as temperate phages, these can remain inactive inside prokaryotic cells for multiple cycles of bacterial replication. This latent phase may end when the lytic cycle is activated, a process that may be triggered by factors such as ultraviolet radiation, temperature changes, or other environmental stimuli.

It is important to note that, unlike the lytic cycle, the lysogenic cycle can be advantageous for the bacterium, since integration of viral genetic material may confer new attributes, such as virulence factors, pathogenic capabilities, or resistance to particular agents. (da Rosa, 2015)

5.3 Pseudolysogenic cycle

Phages are able to replicate only in part of the bacterial population present in the environment, because the remainder of the bacterial population is resistant to their adhesion and genetic-material injection mechanism.

This cycle may be regarded as an intermediate state between the end of the lysogenic cycle and the beginning of the lytic cycle. There is also a type of chronic infection in which the infected bacterium develops resistance to lysis induced by the lytic cycle. In these cases, the release of new phages into the extracellular environment occurs through processes such as budding or extrusion. (da Rosa, 2015)

IV. CHAPTER III

1. Multidrug-resistant bacteria of clinical interest suitable for phage therapy

ESKAPE pathogens are a group of bacteria of particular relevance because of their resistance to more than three groups of antibiotics. The acronym refers to *Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* spp. They represent an ongoing threat to global human health because of their high rates of transmission in both nosocomial and community settings. This imminent health threat has revived interest in the development of new antimicrobial therapies and has highlighted the need for improved patient care.

[Annexes | Figure 8. ESKAPE pathogens] (De Oliveira et al., 2020)

By considering the treatment approaches and fundamental advances that phage therapy has developed for two multidrug-resistant bacteria of major clinical importance, both belonging to the relevant ESKAPE group, we can examine the effectiveness proposed by this method as a viable and crucial alternative.

2. *Staphylococcus aureus*

Staphylococcus aureus is one of the leading agents responsible for disease and death worldwide. This microorganism can cause a broad spectrum of conditions, ranging from relatively mild skin infections to severe conditions such as pneumonia or sepsis. Therapeutic management of these infections is complicated by increasing antimicrobial resistance, and there is currently no effective vaccine against this pathogen.

[Annexes | Figure 9. *Staphylococcus aureus* under the microscope] (Cheung et al., 2021)

2.1 General characteristics

It is a Gram-positive, non-motile, non-spore-forming, facultatively anaerobic bacterium that produces coagulase, is catalase-positive, is capable of beta-haemolysis, is halotolerant, and possesses capsular polysaccharides. Non-selective culture media such as blood agar, chocolate agar, or BHI may be used for isolation.

Likewise, selective culture media such as mannitol salt agar or moderately selective media such as Baird-Parker agar can be used. Colony-forming units (CFU) are observed after 24 hours of incubation. The colonies have a diameter of 0.5 to 1.5 mm. They are also smooth, shiny, creamy, raised, and yellow in colour owing to carotenoid production. (Cervantes-García et al., 2014)

2.2 Resistance

This bacterial species is particularly relevant because of its high degree of virulence and its adaptability to unfavourable conditions. Strains have developed resistance mechanisms against almost all antimicrobials used in treatment. Of particular importance is resistance to the drugs most commonly used to treat infections caused by Gram-positive bacteria, such as beta-lactams, glycopeptides, and oxazolidinones. Most of these cases are caused by methicillin-resistant *Staphylococcus aureus* (MRSA).

Beta-lactam resistance in MRSA strains is associated with transferable genomic islands found in the bacterial genome, known as SCCmec (*staphylococcal chromosomal cassette mec*), in which the *mec* gene determines methicillin resistance. (Hryniewicz, 1999)

Glycopeptides and lipoglycopeptides, like beta-lactams, are bactericidal and act by inhibiting peptidoglycan synthesis, although their precise target and mechanism of action are quite different.

In addition, lipoglycopeptides bind to the bacterial cytoplasmic membrane and cause rapid, concentration-dependent depolarisation of the cytoplasmic membrane, increased permeability, and leakage of cellular ATP and K⁺ ions, leading to cell death. (Zeng et al., 2016). Resistance in vancomycin-resistant *Staphylococcus aureus* (VRSA), particularly at high glycopeptide concentrations, occurs very rarely and is associated with the VanA operon derived from *Enterococcus* spp. through a transduction mechanism. (Courvalin, 2006)

3. *Pseudomonas aeruginosa*

This bacterium is predominant in the hospital environment and causes nosocomial infections, also known as hospital-acquired infections. It is opportunistic, meaning that its pathogenic capacity increases significantly when a patient's defences are compromised, as occurs in chronic disease, immunosuppressive treatment, or previous infections.

[Annexes | Figure 10. *Pseudomonas aeruginosa*] (Roca and Ángel, 2014)

This bacterium is classified by the WHO as a highly dangerous bacterium, previously listed as critical, that must be monitored and controlled because its ability to develop resistance to antibiotics is particularly concerning.

“Carbapenem-resistant *Pseudomonas aeruginosa* (CRPA) has been moved from ‘critical’ to ‘high’ priority in the 2024 update... the high fatal burden of CRPA among immunocompromised individuals and in health-care settings indicate a continued need for innovative R&D”

(WHO Bacterial Priority Pathogens List, 2024)

It should be noted that it has an affinity for moist areas and is highly prevalent in such environments, both outside and within hospitals. This is due to its physiology, metabolism, and adaptive capacity.

3.1 General characteristics

It is a Gram-negative, motile bacterium with a polar flagellum; it is non-spore-forming, strictly aerobic, oxidase- and catalase-positive, capable of beta-haemolysis, halotolerant, and produces pigments such as pyocyanin and pyoverdine. Non-selective culture media such as blood agar, nutrient agar, or BHI can be used for isolation, as well as selective media such as ceftrimide agar.

Strains of this bacterium display a bright green or metallic green colour because of the production of blue pyocyanin and fluorescent yellow pyoverdine. (Paz-Zarza et al., 2019)

3.2 Resistance

It presents multifactorial acquired resistance based on chromosomal mutations and the acquisition of resistance genes through horizontal transfer. These mutations optimise multidrug efflux systems, facilitating the expulsion of drugs from the bacterium. It also has low outer-membrane permeability, which limits the entry of antimicrobials.

The genes acquired can be carried by plasmids, transposons, and prophages. The main mechanisms of gene transfer are conjugation and transduction.

[Annexes | Figure 11. Main mechanisms of gene transfer]

As mentioned above, mutations play an important role in the development of antimicrobial resistance. Examples include alteration of drug target sites, limiting drug uptake, and alteration of membrane permeability due to changes in the function of a porin, thereby rendering the therapy ineffective. (Rocha and Burgos, 2024)

The greatest resistance capacity of *Pseudomonas aeruginosa* is associated with its ability to produce a biofilm capable of protecting it from external agents. These structures provide strong defence and enable adhesion to different hospital surfaces, medical equipment, and both living and inert tissue, even resisting the action of antiseptic substances.

V. RESULTS

1. Positive precedents in the use of bacteriophage therapy

As described above, bacteriophage therapy has been used successfully to treat diseases in which antibiotics have been unable to produce favourable outcomes, either because of bacterial resistance or because access to those drugs was not possible.

Indeed, throughout history, multiple clinical cases with positive and favourable outcomes have been documented for this technique, which is still regarded today as an “exotic” alternative within conventional medicine.

Some of the most relevant precedents in the history of this alternative therapeutic approach are presented below.

1.1 Cystic fibrosis with chronic infections by *Staphylococcus aureus* and/or *Pseudomonas aeruginosa* (2024)

“The Institute for Integrative Systems Biology (CSIC-UV) publishes in the journal ‘Med’ the results of the first phage-therapy treatments using therapeutic vials produced in Spain.”

(CSIC, 2024)

Cystic fibrosis (CF), when combined with an infection caused by a multidrug-resistant bacterium, represents a potentially lethal threat to the patient. In the most severe cases, lung transplantation becomes the most viable therapeutic option for patients affected by this hereditary disease. Following this procedure, there is a high risk of colonisation by multidrug-resistant pathogens such as *Staphylococcus aureus* and/or *Pseudomonas aeruginosa*.

This case, which occurred in Spain, represented a milestone and was carried out through collaboration between the Institute for Integrative Systems Biology (I2SysBio), the Spanish National Research Council (CSIC), and the University of Valencia (UV), together with the Yale Center for Phage Therapy in the United States of America (USA). The treatment was carried out at Hospital Universitario La Fe in Valencia and Hospital Virgen del Rocío in Seville, marking the beginning of the first treatment produced entirely in Spain.

The treatment was structured in three stages in which three different types of phages were used by nebulised administration. (CSIC, 2024)

During the course of treatment, the patient was observed to develop antibodies against the phages. It was also found that the bacteriophages remained active for a period of time in the patient's sputum. One of the main uncertainties was whether the patient might develop an adverse reaction to treatment. Following treatment, it was observed not only that no adverse effects occurred, but also that phage elimination could take place through both the immune system and physiological mechanisms. (Schooley et al., 2017)

1.2 Lung transplant colonised by *Achromobacter xylosoxidans* (2021)

“A Case of Phage Therapy against Pandrug-Resistant *Achromobacter xylosoxidans* in a 12-Year-Old Lung-Transplanted Cystic Fibrosis Patient.”

(Lebeaux et al., 2021)

Unlike the previous case, this publication describes another case of CF involving lung transplantation. Here, colonisation by pandrug-resistant *Achromobacter xylosoxidans* was observed; in other words, a bacterial strain resistant to all clinically available antibiotics, making this a situation of maximum therapeutic urgency.

This degree of resistance is classified as Pandrug-Resistant (PDR), representing the most critical level of bacterial resistance known to date. (Bador et al., 2013)

The treatment was made possible through international collaboration among several institutions, including Hôpital Européen Georges Pompidou (AP-HP) and CHU Lille in France, Queen Astrid Military Hospital and Ghent University in Belgium, the Leibniz Institute DSMZ in Germany, and several French universities including Université de Paris and Paris-Saclay.

Given this extreme situation, phages were selected as the therapeutic strategy. Because of the lack of detailed clinical information and prior data that could support treatment effectiveness, bacteriophage therapy was applied as a last resort.

In this clinical context, two therapeutic strategies were used: phage nebulisation and direct puncture into the lung. Nebulisation was initially selected. However, because there was no clinical improvement, the treatment approach was changed to direct local administration. Long-term follow-up then showed a notable improvement, reinforcing the value of phage therapy as an alternative in the face of bacterial antibiotic resistance. (Lebeaux et al., 2021)

1.3 Necrotising pancreatitis complicated by a pancreatic pseudocyst infected with multidrug-resistant *Acinetobacter baumannii* (2017)

“Development and Use of Personalized Bacteriophage-Based Therapeutic Cocktails to Treat a Patient with a Disseminated Resistant *Acinetobacter baumannii* Infection.”

(Schooley et al., 2017)

Continuing with the cases presented, this section describes the intravenous and percutaneous administration into abscess cavities of a phage cocktail. The trigger for this therapy was *A. baumannii*, a pathogen listed among the ESKAPE organisms mentioned previously. Like other members of the ESKAPE group, *A. baumannii* has an important reservoir in the hospital environment.

Collaboration among several institutions was essential for this treatment. The patient was treated at UC San Diego Health, while development and analysis of the phage cocktail were carried out by the Center for Phage Technology at Texas A&M University and the US Navy Medical Research Center. In addition, AmpliPhi Biosciences contributed purified phages for compassionate use, allowing intravenous administration to be combined with intra-abdominal administration.

What generated the greatest clinical interest in this case was the speed of recovery observed after direct treatment was applied both intravenously and directly to the affected area. At the same time, the pathogen developed significant sensitivity to antibiotics following treatment. The use of phage therapy in this context was primarily justified by the high prevalence of this pathogen among US military personnel deployed in conflict zones. (Schooley et al., 2017)

1.4 Pioneering historical cases

1.4.1 Treatment of dysentery (1919)

Dr D'Hérelle applied bacteriophage therapy for the first time in a patient with dysentery, caused by *Shigella* spp. This case marked the beginning of a therapeutic revolution in clinical medicine that was later overshadowed by antibiotics.

He thereby demonstrated that bacteriophage therapy was a safe and effective treatment without adverse effects. The patient experienced a rapid and complete recovery within 24 hours. The phage was administered orally. (Phage 101 - IPATH, n.d.) (d'Herelle, 1922)

1.4.2 Large-scale bacteriophage-therapy programmes in the USSR (1930s)

In 1934, D'Hérelle accepted an invitation to the Soviet Union (USSR), where he worked alongside his colleague and friend Professor George Eliava - who had founded the Tbilisi Bacteriophage Institute - to promote and develop different phage treatments against multiple pathogens. The primary targets included cholera, dysentery, typhoid fever, and other common infections of the period.

Their joint work was highly successful, allowing large-scale epidemic outbreaks to be controlled and leaving a notable legacy in the former Soviet Union. (Phage therapy | Summary, n.d.)

2. Clinically relevant aspects of phage therapy

Following analysis of several clinical cases involving the use of this alternative therapy, it has been observed both in those cases and in published data that no adverse effects on patient health have been identified in relation to toxicity or immunosuppression, which are among the potential adverse effects commonly monitored during clinical therapy.

It is particularly important to emphasise that the human microbiota is not composed only of bacteria, but also contains bacteriophages, viruses, and fungi.

[Annexes | Figure 12. Interaction among the three kingdoms of the microbiome] (Popescu et al., 2021)

The most relevant aspects from a clinical point of view are addressed below, classifying the effects observed in current applications.

2.1 Advantages

2.1.1 High specificity

Phages have a strong ability to recognise and selectively attack a specific bacterial strain without significantly affecting other microorganisms. Unlike many broad-spectrum antibiotics, this specificity makes them highly innovative as a potential alternative therapy. (Cui et al., 2024)

2.1.2 Demonstrated clinical effectiveness

Available documentation from different clinical cases records that the therapy works both independently and in combination with antibiotics. It has also shown not only that it can combat pathogens, but also that it may help mitigate the negative effects of certain diseases or host infections.

For example, a study performed in a murine model of collagen-induced arthritis showed that administration of T4 phages significantly reduced disease severity. Treated mice showed reduced pro-inflammatory cytokines, reduced T-cell activity, and increased anti-inflammatory factors. (de Souza et al., 2023)

Likewise, one of the major potential benefits of this therapeutic strategy lies in its capacity to reduce bacterial resistance and its development against antibiotics when the two are used in combination. This synergy further reinforces its clinical value in the current context of the antimicrobial-resistance crisis. (Cui et al., 2024) (CIDRAP, 2024)

2.1.3 Low or no toxicity and self-limiting effect

Phages have not shown significant toxic or immunosuppressive effects in patients, even in prolonged treatments. One factor that supports their safety profile is that, after administration, the phage remains active only against the target pathogen, without affecting other cells or tissues in the body.

A clinical trial conducted in 2019 showed that, after administration of an oral cocktail containing four types of phage directed against *E. coli*, the population of this bacterium decreased significantly without altering the overall composition of the remainder of the patient's microbiota. (Divya Ganeshan and Hosseinidoust, 2019)

2.2 Disadvantages

2.2.1 Possibility of pathogen recurrence and possible development of phage resistance

One disadvantage that must be considered is the possibility that the pathogen treated with phage therapy - particularly when the therapy is used as the only treatment strategy - may reappear. This phenomenon may result from bacterial evasion mechanisms or incomplete elimination of the pathogen, which can compromise long-term effectiveness.

Nevertheless, this does not mean that the therapy is ineffective on its own; rather, as occurs with antibiotics, pathogens may develop a degree of resistance or adaptive capacity against phages. (CIDRAP, 2024)

Although this represents a clinical disadvantage, just as the pathogen is capable of adapting to the phage, the phage is also capable of adapting to the pathogen. This adaptation allows the phage to evolve and identify new strategies for neutralising the pathogen.

2.2.2 Possible development of resistance to the phage by the patient

The use of phages generates both innate and adaptive immune responses in the individual, because they are recognised as foreign agents. Their presence therefore activates the patient's immune response, indicating that phages can modulate different components of the immune system.

Following prolonged exposure - approximately two or three weeks - the immune system has been observed to develop antibodies against the administered phages. Although this is beneficial in the sense that it does not involve an adverse effect for the patient, it may slow the treatment process.

There is evidence that antibody formation occurs more frequently when the patient is treated directly, that is, through intravenous administration or local administration at the affected site. (Popescu et al., 2021)

This does not imply that the organism is harmed by administration of the therapy. Rather, as with anything foreign to the body, phages are recognised as foreign elements. Different clinical and experimental cases have shown that, when phages are used, the immune system generates both pro-inflammatory and anti-inflammatory responses, demonstrating potentially beneficial immunomodulatory capacity. (Popescu et al., 2021) (de Souza et al., 2023)

2.3 Current barriers

2.3.1 Limited access and lack of regulated availability

In many countries, phage therapy is not approved as a conventional treatment, limiting its availability outside research settings and compassionate-use cases.

This, together with the absence of clear international regulations for the authorisation, control, and distribution of clinical phages, slows implementation in hospitals and public healthcare centres.

2.3.2 Personalised production and costs

Phage-therapy treatment requires identification of a phage with specific affinity for the pathogen causing the infection, making it necessary to design personalised treatments for each patient.

The resources required and the production time increase considerably, thereby raising treatment costs compared with broad-spectrum antibiotic treatment. From an economic rather than clinical perspective, this represents a limitation in terms of cost-effectiveness compared with conventional treatments.

3. Processes for phage inoculation in cultures

In nature, phages and bacteria maintain a dynamic co-evolutionary relationship. Although bacteria may develop defences against phages, these viruses are also able to evolve and overcome those barriers, recovering their capacity to infect them. For this reason, they are regarded as “adaptive medicines”. In contrast, antibiotics are stable chemical compounds that cannot alter their structure in response to new forms of bacterial resistance. Phages, by contrast, are highly versatile and may be used in several ways: individually, in combination with other phages or antibiotics, or through enzymes derived from them. (Wittebole et al., 2014)

3.1 Application and phage process

The first requirement for applying phage therapy is to have bacteriophages suitable for treatment, which remains difficult because of the current stage of development of these viruses. It is necessary to have phages that are active against the patient's specific bacterium and that also meet regulatory standards such as purity, traceability, and characterisation. Phages may be used individually (monophage preparation) or several may be combined in a cocktail to target one or more bacterial species.

Because phage therapy is not yet approved as a medical treatment in Western countries and there are no officially registered products, phages must be prepared specifically for each infection (personalised therapy). Alternatively, commercially prepared products from Russia or Georgia may be used, as these already contain defined phage combinations and are ready for use. (Pirnay et al., 2011)

3.2 Identification of the bacterial strain

A laboratory study is carried out using cultures of different strains in order to identify the strain responsible for the symptoms in affected patients. These procedures use techniques such as microbiological culture, cryopreservation, and different types of staining, including Gram staining, in order to identify, preserve, and handle samples with the greatest possible care, supporting patient treatment by producing more reliable and accurate results. (Roiter Jaimes, 2023)

3.3 Phage isolation

As with identification of the pathogen, the phage must be highly specific in order to support treatment viability without affecting the patient's own microbiota. Therefore, some steps in the selection of phage treatment - which may complement antibiotics or other techniques - focus on isolating phages collected from the hospital environment or obtained on request from specialised centres. (Patey et al., 2019) Traceability must always be maintained so that their function can be properly fulfilled. Storage takes place in freezers at -20°C.

[Annexes | Figure 13. Steps for preparing a phage suspension] (Roiter Jaimes, 2023) (Vázquez et al., 2022)

3.4 Phage administration and application

Depending on the active phage component and the route of administration with the greatest therapeutic potential, the method of administration is selected from among the following options.

3.4.1 Topical application

Phages may be used topically in creams or gels, or applied through moist dressings placed over the wound. Because bacteriophages are composed of proteins, there is some concern regarding possible anaphylactic reactions with repeated use. Nevertheless, severe reactions have been uncommon, and the risk is currently even lower thanks to modern purification techniques. (Międzybrodzki et al., 2012) (Jault et al., 2019)

3.4.2 Oral administration

Although administration is relatively straightforward and does not cause side effects, stomach acidity can destroy phages. To avoid this, gastric pH can be neutralised using alkaline liquids (such as water with bicarbonate), or the phages can be protected in acid-resistant capsules. However, neutralisation may increase the risk of opportunistic infections, while the use of capsules increases costs. More clinical studies are still required to determine the best option for oral administration. (Międzybrodzki et al., 2012)

3.4.3 Inhalation

This is a promising route for administering lyophilised bacteriophages to the lungs as powder through inhalers. Because the respiratory system is accessible through the airways, phages may also be delivered as a suspension (nebulisation) or as a dry aerosol. However, there are still few published studies supporting this route of administration. (Abedon, 2015)

3.4.4 Intravenous administration

Bacteriophages are distributed rapidly, but are captured by the reticuloendothelial system, particularly in the spleen and liver. If there are no target bacteria, they are rapidly eliminated. By contrast, if susceptible bacteria are present, the phages replicate, although the amount depends on factors associated with both the bacteria and the host organism, making their behaviour difficult to predict in each patient. (Abedon, 2014)

VI. DISCUSSION

The clinical cases reviewed, together with the evidence collected within this theoretical framework, provide a solid basis for reflecting on this treatment with considerable potential, whose prominence has increased substantially in recent years.

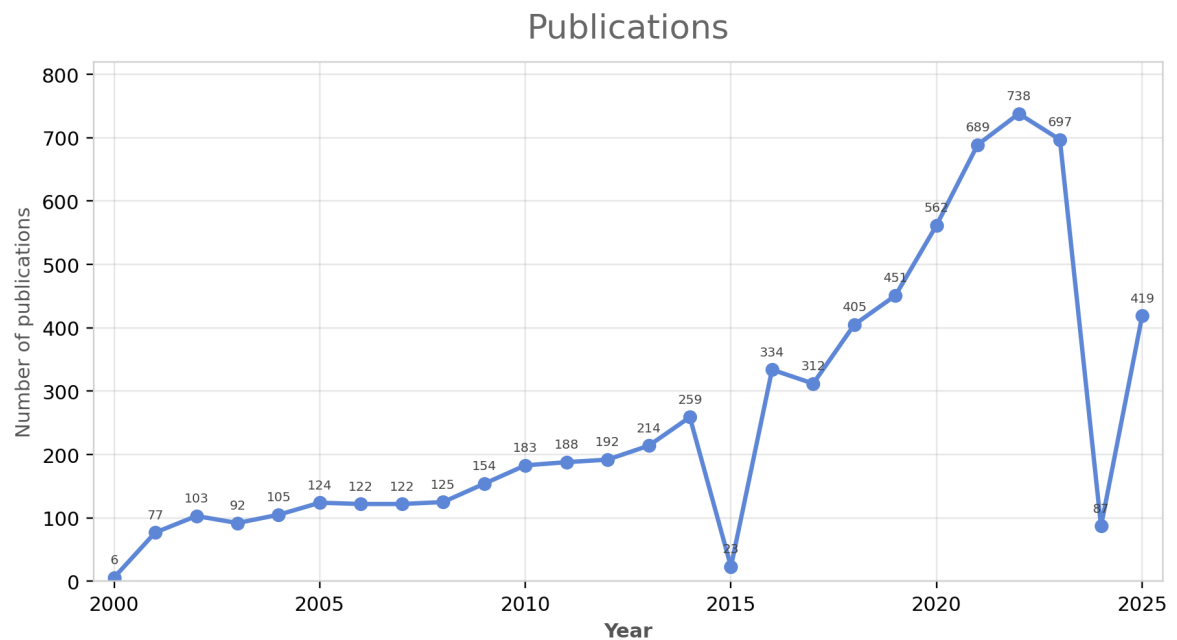


Chart 1. Evolution of scientific publications on phage therapy (PubMed, 2000-2024)

Source: Authors' own work based on data obtained from PubMed (search: "phage therapy")

The literature review demonstrates that phage therapy, both as an independent treatment and in combination with other treatments, produces positive results against infections caused by bacteria. Several studies reinforce this observation, showing how diseases caused primarily by ESKAPE pathogens have been treated. (Lebeaux et al., 2021)

In addition, several studies have observed that phage therapy has the capacity to help regulate inflammatory processes, whether through direct action of the pathogen or through other effects affecting the organism. (Cui et al., 2024) (de Souza et al., 2023)

Despite all the favourable aspects presented by this innovative treatment, its full clinical approval remains under development. Partial uncertainty regarding certain mechanisms of action, together with the limited number of large and standardised clinical studies, has

relegated phage therapy to the status of a last-resort option, even though in different situations it has had to be used when everything else failed, whether as a complementary or independent treatment. (Wittebole et al., 2014)

This contradiction raises the important question of whether the full potential of phages is really being recognised as safe therapeutic agents with low or no toxicity for the body, in contrast with the adverse effects associated with many antibiotics.

The WHO has already warned that we are entering a post-antibiotic era, in which common infections may become untreatable. Given this situation, in which there is a therapy with such potential alongside a scenario marked by a growing antibiotic crisis, the need to study phage therapy in greater clinical depth is more than evident, particularly with regard to its standardisation and regulatory approval. With so many clinical cases and preliminary trials, the evidence is more than clear: the therapy works.

For this progress to continue in a reasoned and intelligent manner, it will be essential to foster synergy among scientific research, hospitals, and the pharmaceutical industry. In this way, this major contribution made so many years ago may finally receive the recognition it deserves as an alternative, safe, effective, and necessary therapy, and thus become established as a therapeutic alternative for the twenty-first century.

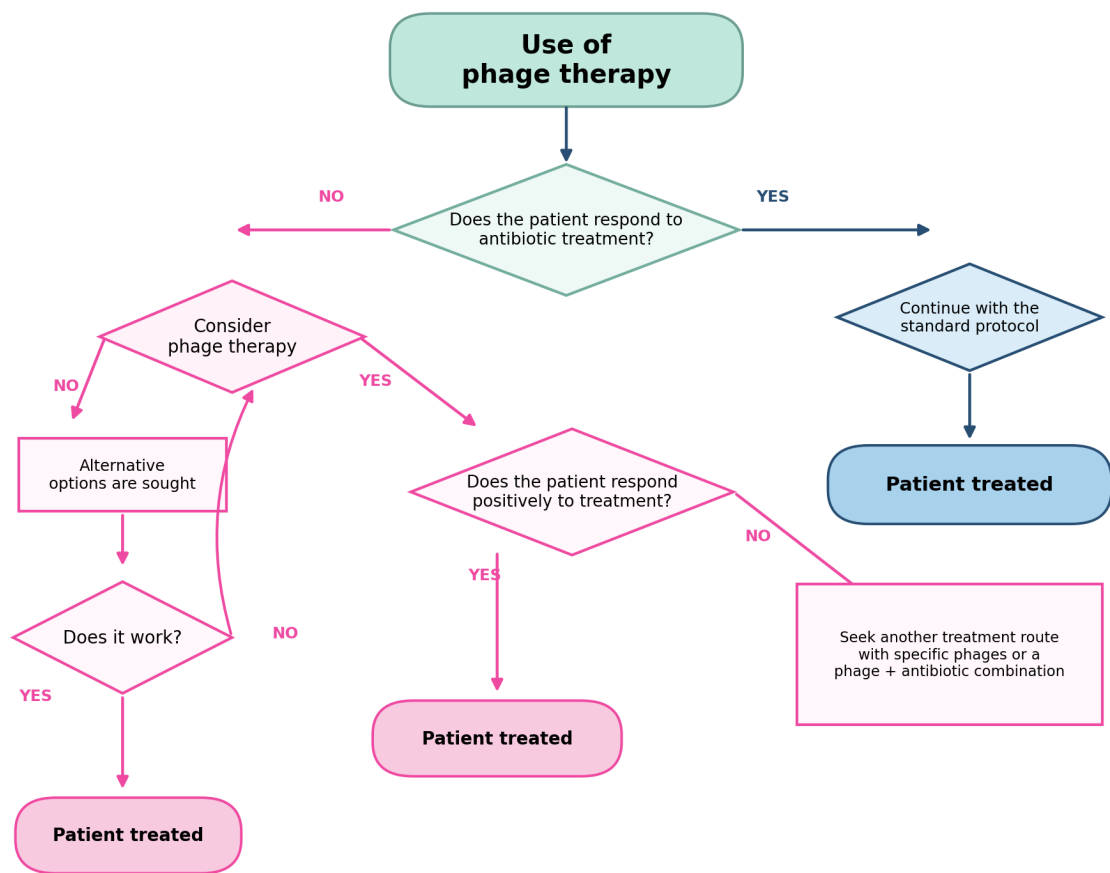


Figure 2. Flowchart on the use of phage therapy

Source: Authors' own work

VII. CONCLUSIONS

Bacteriophages represent a promising therapeutic tool in the fight against bacterial infections, particularly those caused by strains that are multidrug-resistant to antibiotics. Their high specificity, which allows pathogenic bacteria to be selectively targeted without damaging beneficial microbiota, makes them a particularly valuable alternative to conventional treatments. In addition, their capacity to co-evolve with bacteria gives them a dynamic advantage in the context of antimicrobial resistance, opening the door to more personalised and adaptive treatments.

The potential of phages is expanded when they are combined with other therapies, such as antibiotics or lytic enzymes derived from the viruses themselves, which may enhance treatment effectiveness and reduce the emergence of resistance. However, despite scientific and clinical advances, the main obstacle to widespread implementation remains the absence of a specific legal framework that would allow their use according to established protocols. At present, their application is largely limited to compassionate or experimental treatments, restricting their reach within the healthcare system.

It is therefore essential for health authorities and regulatory bodies to work on the development of clear regulations adapted to the particular characteristics of phage therapy. Only then will it be possible to integrate this innovative strategy into routine clinical practice, allowing its high therapeutic potential to translate into real benefits for patients.

1. Hypothesis on the therapeutic potential of phage therapy in combination with other antimicrobial strategies

Based on the firm and promising foundations identified within phage therapy, we can envision some of the directions in which it may develop. The most important function of phages in treatment is their lytic action, from which their various characteristics can be further developed. In a context of growing bacterial resistance, these new approaches are particularly relevant.

1.1 Lytic phages as a base therapy

The direct use of lytic bacteriophages remains the foundation of this strategy. These viruses specifically attack pathogenic bacteria without affecting the microbiota or human cells, making them a safe and targeted alternative. Their effectiveness depends on the phage selected for the type of infection.

1.2 Phage enzymes as antimicrobial agents

Among the derived tools, endolysins stand out: enzymes that destroy the bacterial cell wall. These proteins may be used as a direct treatment without requiring the complete virus and have shown particular effectiveness against Gram-positive bacteria. They also present a low probability of generating resistance.

1.3 Genetically modified phages

Genetic engineering has made it possible to design more efficient phages, broaden their range of action, and, in some cases, turn them into vehicles for gene therapies, such as CRISPR systems directed against resistance genes. This biotechnological approach opens a highly innovative route in the fight against multidrug-resistant bacteria. (Abedon, 2014)

1.4 Phages + antibiotics: therapeutic synergy

One of the most promising strategies is the combination of phage therapy with antibiotics. It has been shown that phages can facilitate the action of antibiotics by weakening bacterial defences and, in some cases, even forcing bacteria to lose their resistance. This synergy not only improves treatment effectiveness, but may also reduce the antibiotic doses required, thereby decreasing side effects and selective pressure. (Cruz Díaz et al., 2025)

1.5 New biotechnological combinations

Finally, phage therapy can also be integrated with other modern tools such as probiotics, nanoparticles, and genetic technologies. These combinations allow a more precise approach adapted to the type of infection, the patient, and the clinical environment, contributing to more personalised medicine. This is an especially important point in relation to phages because of their demonstrated specificity. (Cruz Díaz et al., 2025)

In conclusion, phage therapy should not be understood solely as an independent solution, but as an expanding therapeutic platform capable of being combined

with antibiotics, biotechnology, and products derived from phages themselves. Its versatility and specificity make it a tool with major prospects for the future treatment of bacterial infections.

These reflections are grounded in the continuing advances that have been developed through phage research. This section therefore represents a future-oriented view of the direction the healthcare sector may take as its main weapon against microbial infections becomes compromised.

Accordingly, the hypothesis presented here reflects an assumption regarding the path that I consider appropriate to pursue and support. This text is solely a composition by Luis Gonzalo Legua Pérez, whose perspective complements the bibliographic work on which we collaborated.

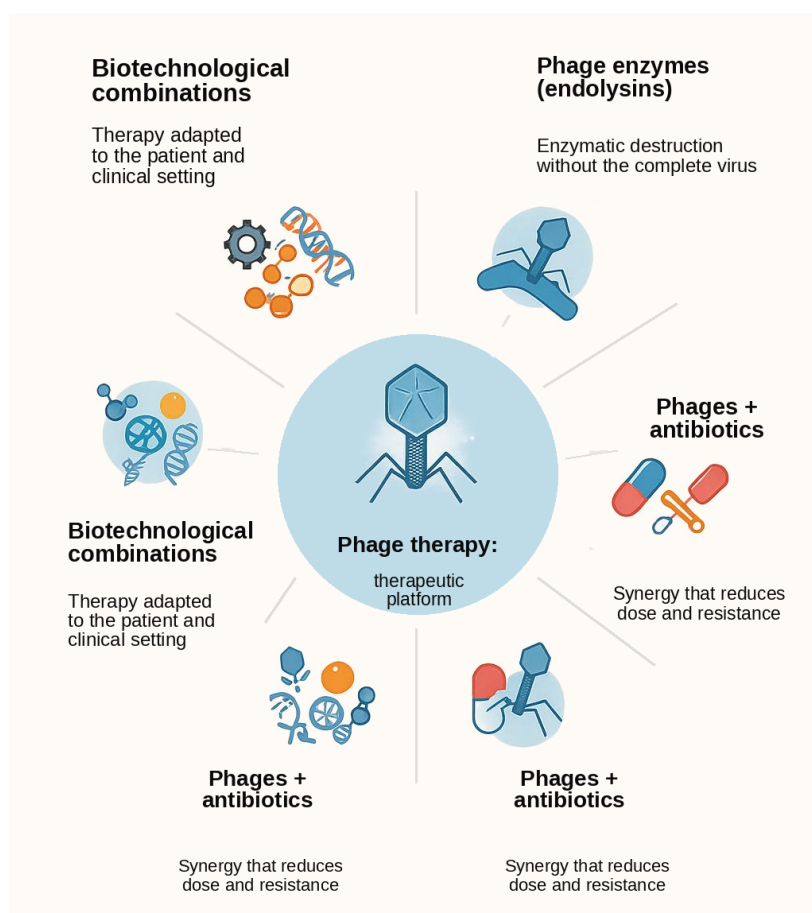


Figure 3. Therapeutic projection of phage therapy in combination with antimicrobial technologies

Source: Conceptual composition by Luis Gonzalo Legua Pérez, as a hypothesis for future projection based on the bibliographic study conducted

2. Interdisciplinary hypothesis | combination of phage therapy and chitosan for the elimination of microplastics

One of the most relevant findings gathered in this project is the capacity of certain phages to exert anti-inflammatory and immune-regulatory effects. Although phage therapy has traditionally been regarded exclusively as a treatment for eliminating pathogenic bacteria, these observed immunomodulatory effects open new avenues for research and new treatments.

This positions phage therapy as a potentially useful strategy in the management of inflammatory, immune-mediated, or even chronic diseases in which modulation of the immune system represents a priority therapeutic objective. In this way, its use could extend from bacterial control to the design of combined therapies with regulatory effects on the patient.

In this context, it is relevant to consider that the elimination of microplastics from the body could have a direct impact on the treatment of certain inflammatory or immune-mediated diseases. The presence of microplastics in the body has been associated with alterations in the microbiota, chronic inflammatory processes, metabolic disruption, pulmonary diseases, and even diseases affecting the nervous system (Liu and Shimizu, 2025). Their elimination through innovative therapeutic strategies could therefore contribute to restoring the host's immune, metabolic, and neurological balance.

Based on analysis of the immunological and microbiological effects of phage therapy, together with study of the potential of chitosan as an adsorbent biopolymer, an interdisciplinary hypothesis is proposed that combines both strategies for advanced therapeutic purposes. Chitosan has demonstrated a high capacity for microplastic adsorption and high biocompatibility with biological organisms. (Nicholas, 2024) (Prasetyo et al., 2025)

Once in the body, these microplastics may be covered by bacterial biofilms that make their elimination more difficult and could generate adverse immunological or metabolic effects. It is proposed that, by combining chitosan with specific phages directed against the bacteria colonising these biofilms, it would be possible to destabilise the

microplastic-bacterial complex. This action would favour the dissolution or release of the microplastic, facilitating its subsequent elimination from the body through physiological routes such as urine or faeces, with the assistance of the chitosan-microplastic complex formed. (Korniienko et al., 2024)

In addition, high-throughput sequencing would make it possible to identify microorganisms with a natural affinity for microplastics, facilitating the selection or design of highly specific phages for this combination.

This hypothesis does not propose that phages act directly on the plastic polymer, because their action is limited to bacterial organisms. Rather, it proposes an indirect route of intervention based on the interaction among microbiota, microplastics, and functional biomaterials.

To date, no published scientific evidence has been identified that combines phage therapy with the use of chitosan for the elimination of microplastics from the human body. Therefore, this hypothesis constitutes an original and individual contribution by Jhon Michael Cuenca Jaramillo within the context of the present collaborative project.

This proposal is based on knowledge already described in the scientific literature and cited previously throughout the development of the project.

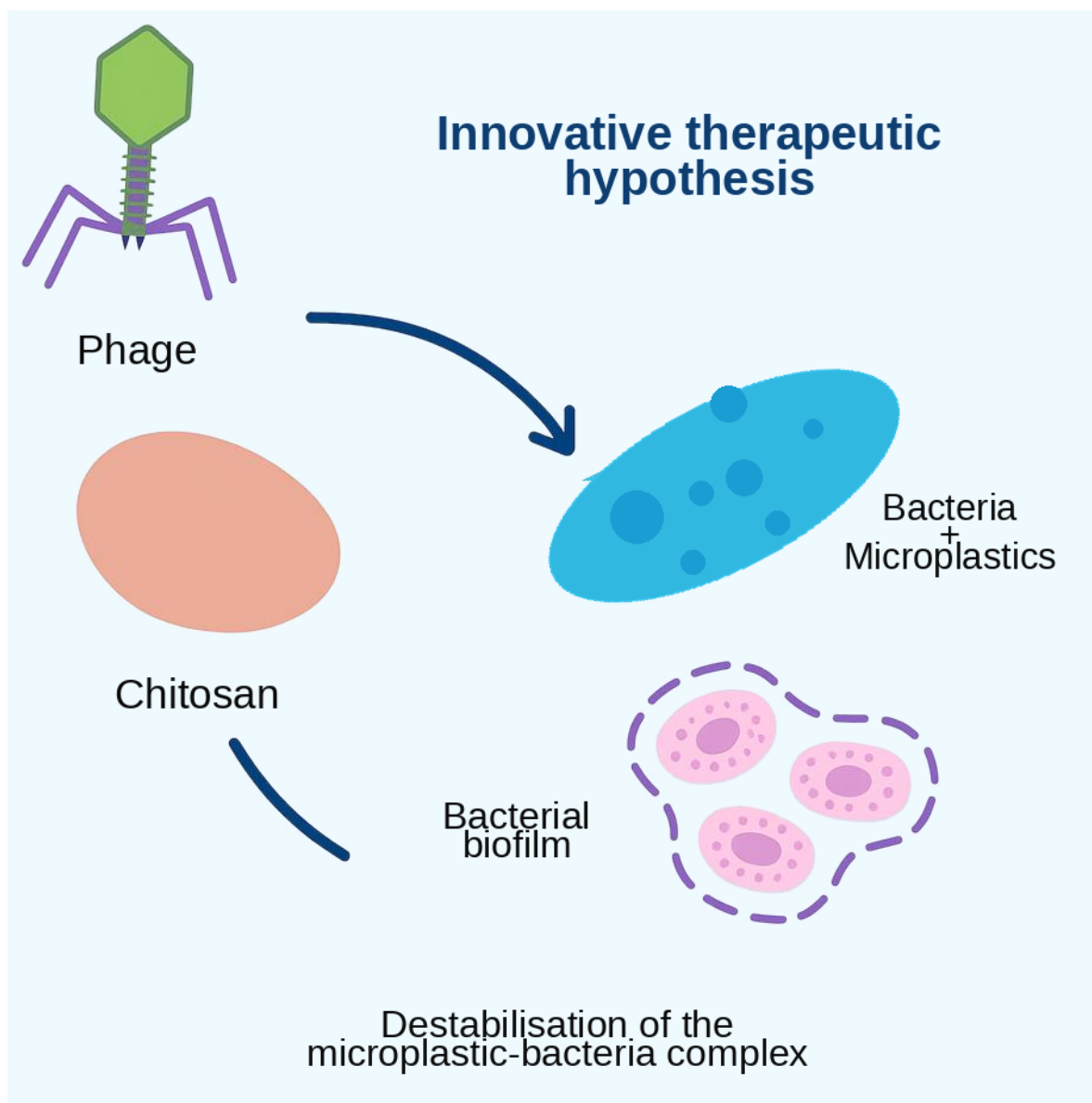


Figure 4. Schematic representation of the proposed therapeutic hypothesis for the elimination of microplastics using chitosan and phage therapy.

Source: Authors' own work based on the hypothesis developed by Jhon Michael Cuenca Jaramillo.

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GLOSSARY OF TERMS

A

Antibiotic	Chemical substance used to inhibit or eliminate bacteria, but whose excessive use has contributed to the increase in antimicrobial resistance.
Antimicrobial resistance	The ability of microorganisms to survive and multiply despite the use of antimicrobial medicines intended to eliminate them.

B

Bacteriophage	A virus that infects bacteria exclusively. Bacteriophages are used therapeutically because of their specificity and lytic capacity.
Bacteriophage therapy	A more technical term for phage therapy. It refers directly to the therapeutic use of bacteriophages.
Bacterial lysis	The process by which a bacterial cell ruptures and dies, commonly caused by the action of bacteriophages or antibiotics.

C

Chitosan	A natural biopolymer with the capacity to adsorb microplastics and used for therapeutic purposes because of its biocompatibility.
Chronic infections	Infections that persist over time, may recur, and can be difficult to treat with conventional therapies.
Combination therapy	The combined use of phage therapy and antibiotics to increase treatment effectiveness against difficult infections.
Compassionate use	Use of an experimental treatment for a patient when no other therapeutic options are available.

E

Efflux pumps	Proteins that expel unwanted substances from bacteria, such as antimicrobials and toxins.
Endolysins	Enzymes derived from bacteriophages that degrade the bacterial cell wall and are used as antimicrobial agents.

G

Genetic engineering	A branch of biotechnology that allows an organism's genetic material to be modified in order to obtain new characteristics or functions.
Genetically modified phage	A bacteriophage altered through genetic engineering to improve its specificity or therapeutic effectiveness.

H

High-throughput sequencing	A genetic-analysis technique that allows microorganisms and their genes to be identified by reading large volumes of DNA.
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I

Immunomodulation	The capacity of a substance or treatment to modify or regulate the immune system.
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L

Lytic cycle	Reproductive cycle of bacteriophages in which the virus destroys the bacterial cell after replicating inside it.
Lytic phage	A type of bacteriophage that infects and destroys bacteria through the lytic cycle.

M

Microbiota	The community of microorganisms that inhabit an organism, such as the human gut, and that perform key functions in health.
Microplastics	Small plastic particles that may be found in the human body and are associated with negative health effects.
Multidrug-resistant bacteria	Bacteria that are resistant to multiple types of antibiotics, making clinical treatment more difficult.

P

Phage enzymes	Molecules produced by bacteriophages that help destroy bacteria, such as endolysins.
Phage therapy	Treatment that uses bacteriophages to eliminate pathogenic bacteria, particularly in antibiotic-resistant infections.
Porins	Proteins that form channels in cell membranes and allow small molecules to pass through.

S

Specificity	The ability of a bacteriophage to attack only a particular bacterial species or strain without affecting other cells in the organism.
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T

Tropism	The selective affinity that an agent, such as a virus, has for a specific type of cell or organism.
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V

Virulence

The ability of a pathogenic microorganism to cause harm to its host.

ANNEXES

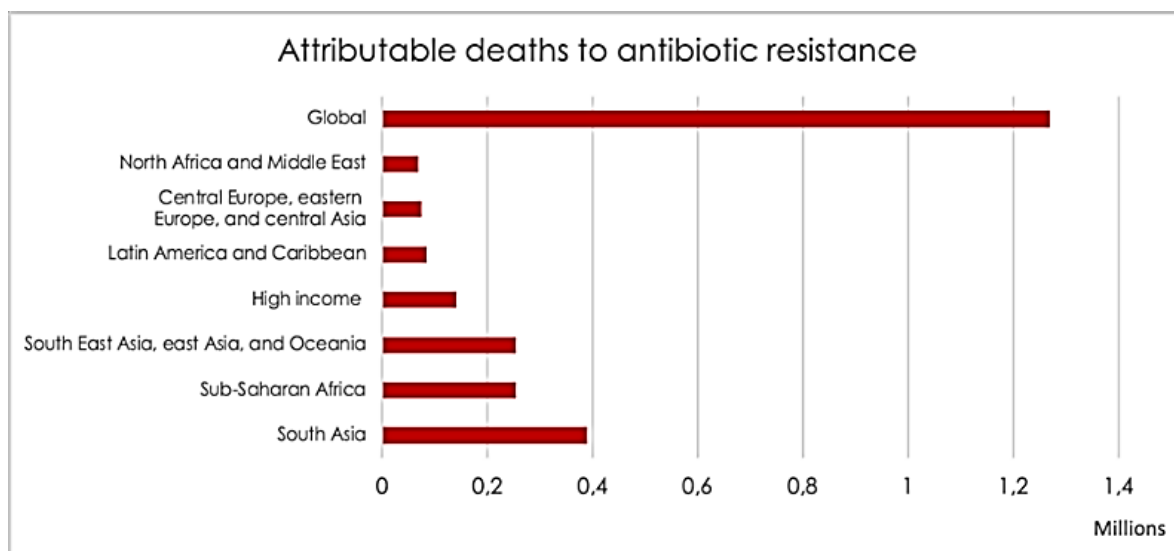


Figure 5. Deaths attributable to antibiotic resistance (GRAM Study)

Source: (Salud y Fármacos, 2022)

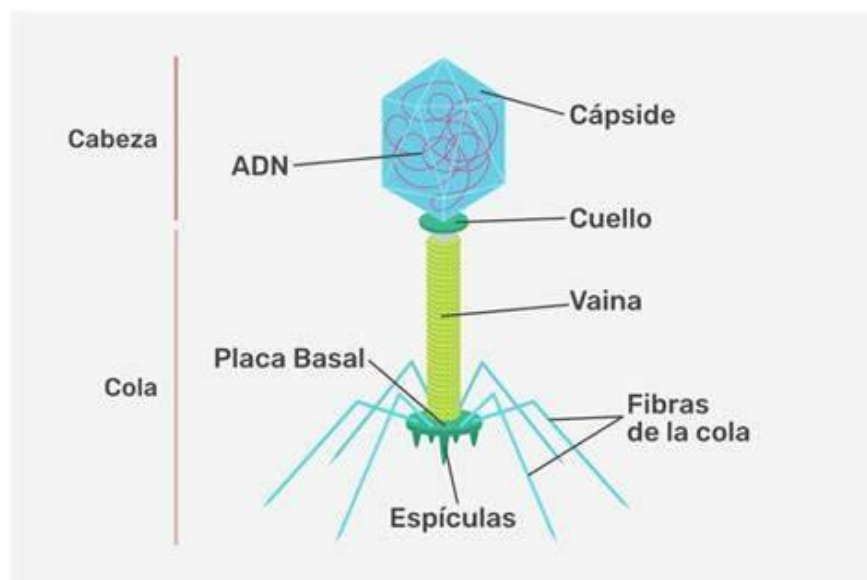


Figure 6. Bacteriophage

Source: (Red de Recursos Educativos en Abierto, n.d.)

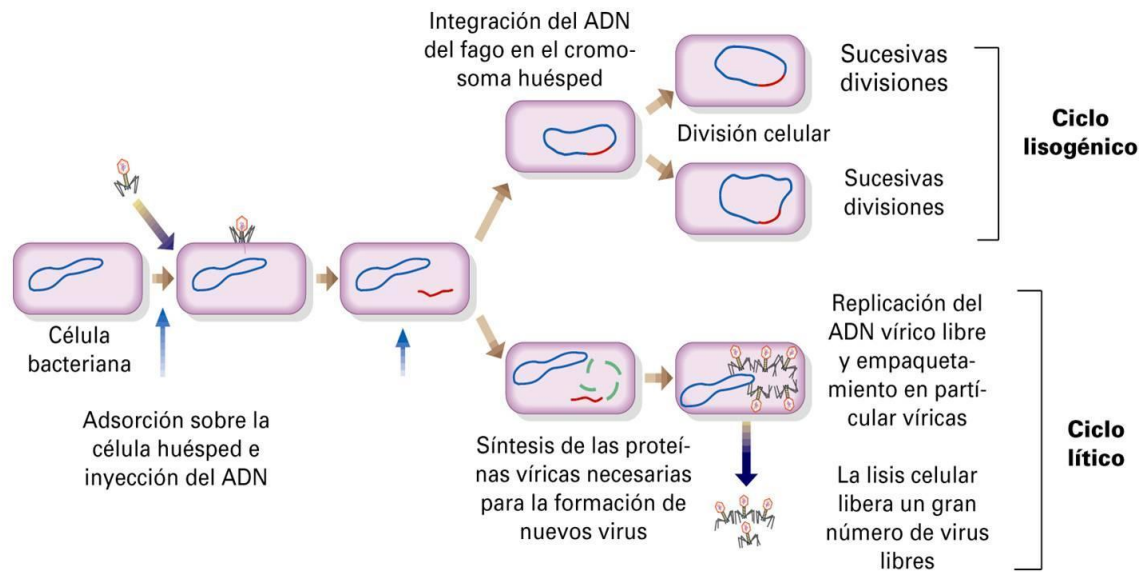


Figure 7. Phage cycle

Source: (Melendez, n.d.)



Figure 8. ESKAPE pathogens

Source: (Clover Bioanalytical Software, n.d.)

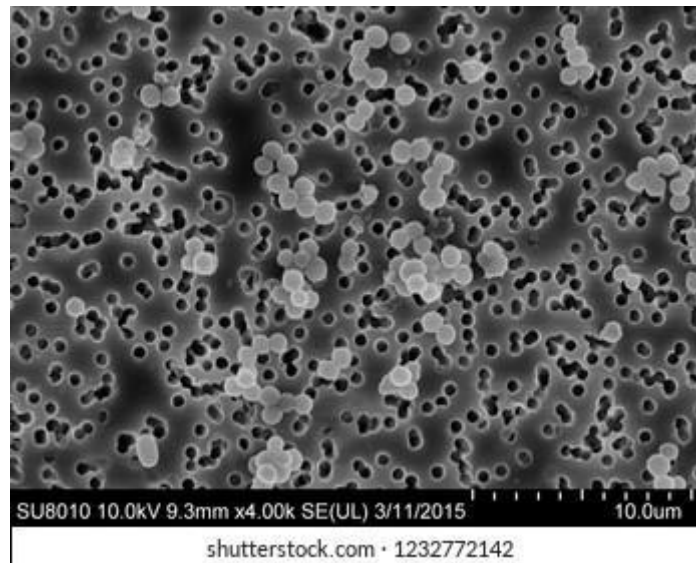


Figure 9. *Staphylococcus aureus* under the microscope

Source: (Stock Photo - Image of hospital, n.d.)

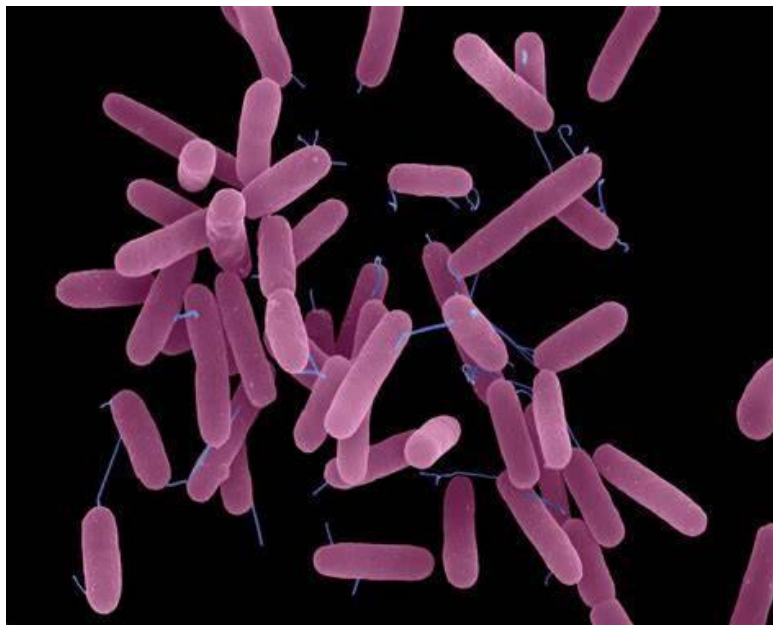


Figure 10. *Pseudomonas aeruginosa*

Source: (Kunnel, n.d.)

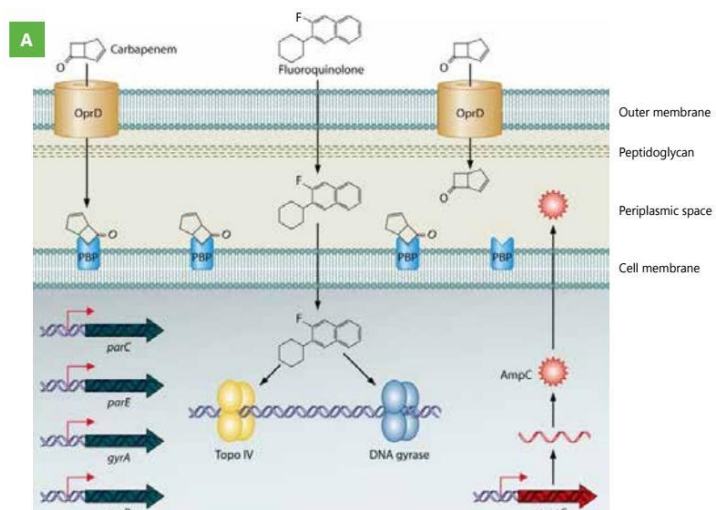


Figure 11. Main mechanisms of gene transfer

Source: (Rocha and Burgos, 2024)

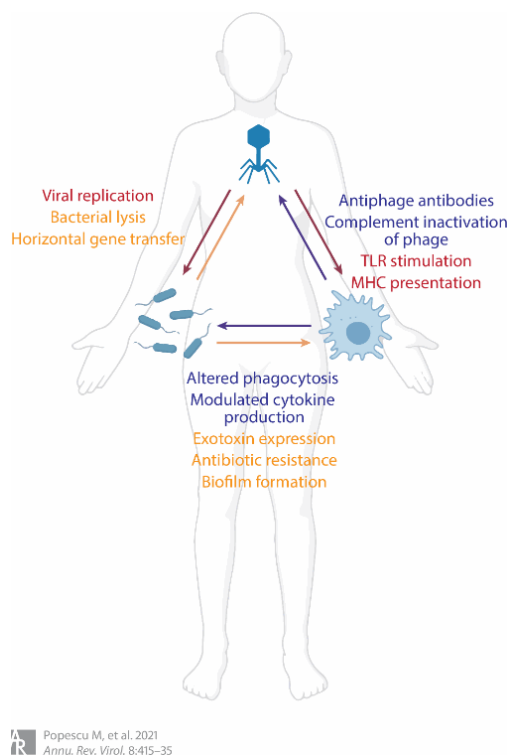


Figure 12. Interaction among the three kingdoms of the microbiome

Source: (Popescu et al., 2021)

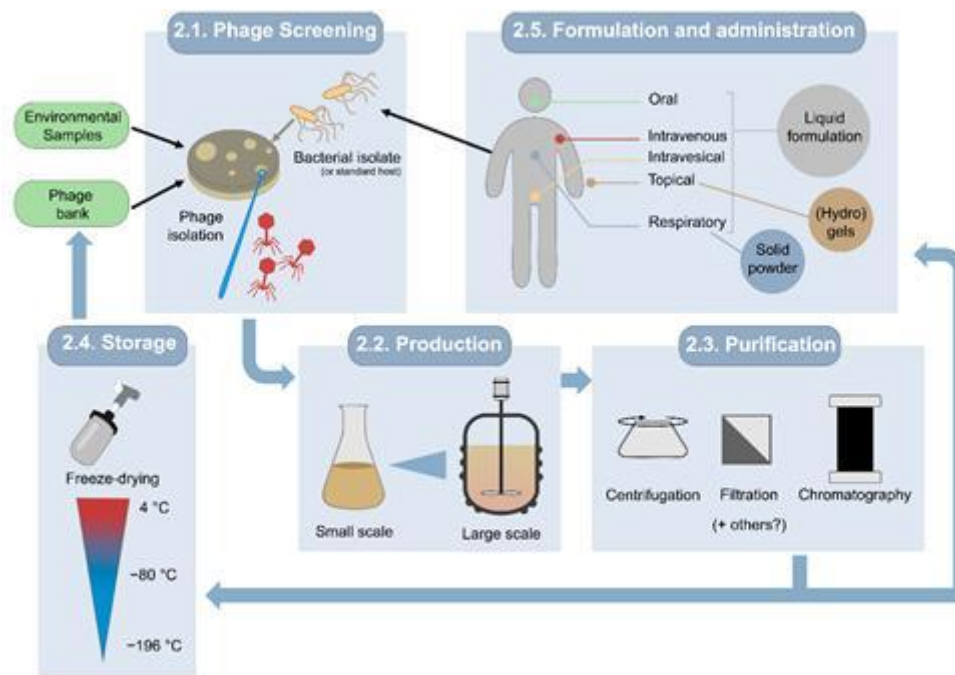


Figure 13. Steps for preparing a phage suspension

Source: (Vázquez et al., 2022)